Basic Data Structures: Stacks and Queues

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Data Structures

Data Structures and Algorithms

Outline

1. Stacks
2. Queues

LIFO = Last In - First Out

Definition

Stack: Abstract data type with the following operations:

- `Push(key)`: adds key to collection
- `key Top()`: returns most recently-added key
- `key Pop()`: removes and returns most recently-added key
- `Boolean Empty()`: are there any elements?

Stack Implementation with Array

numElements: 0

Stack Implementation with Array

numElements: 0

Push(a)

Stack Implementation with Array

numElements: 1

Push(a)

Stack Implementation with Array

numElements: 1

Stack Implementation with Array

numElements: 1

Push(b)

Stack Implementation with Array

numElements: 2

Push(b)

Stack Implementation with Array

numElements: 2

Stack Implementation with Array

numElements: 2

Top()

Stack Implementation with Array

numElements: 2

$\text{Top}() \rightarrow \text{b}$

Stack Implementation with Array

numElements: 2

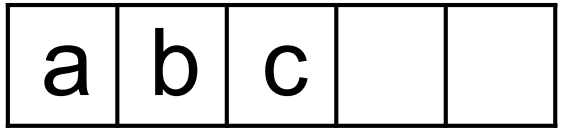
Stack Implementation with Array

numElements: 2

Push(c)

Stack Implementation with Array

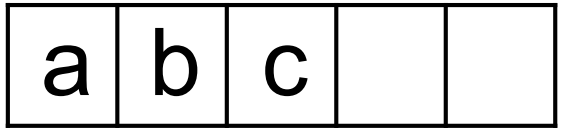
numElements: 3



Push(c)

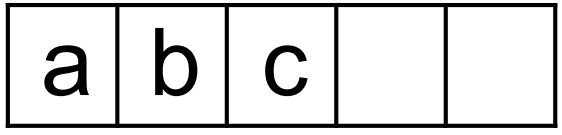
Stack Implementation with Array

numElements: 3



Stack Implementation with Array

numElements: 3



Pop()

Stack Implementation with Array

numElements: 2

$\text{Pop}() \rightarrow \text{c}$

Stack Implementation with Array

numElements: 2

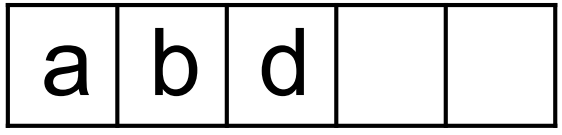
Stack Implementation with Array

numElements: 2

Push(d)

Stack Implementation with Array

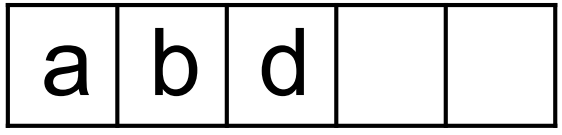
numElements: 3



Push(d)

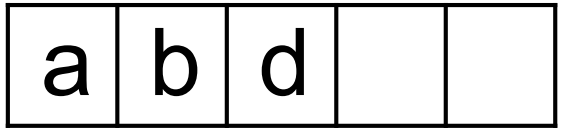
Stack Implementation with Array

numElements: 3



Stack Implementation with Array

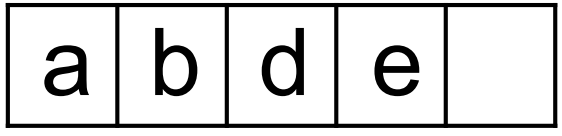
numElements: 3



Push(e)

Stack Implementation with Array

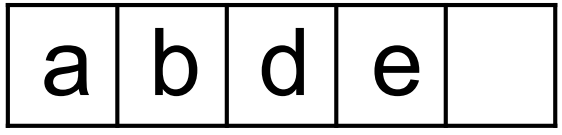
numElements: 4



Push(e)

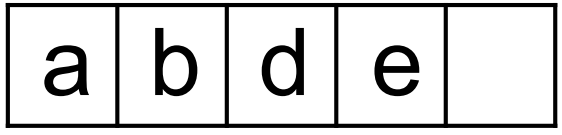
Stack Implementation with Array

numElements: 4



Stack Implementation with Array

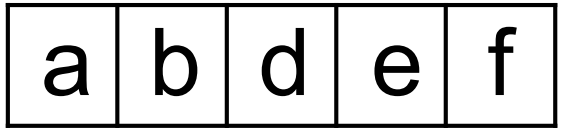
numElements: 4



Push(f)

Stack Implementation with Array

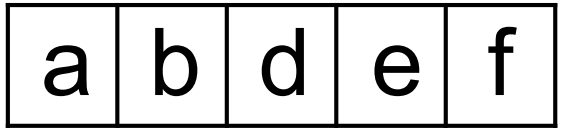
numElements: 5



Push(f)

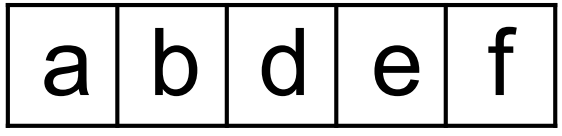
Stack Implementation with Array

numElements: 5



Stack Implementation with Array

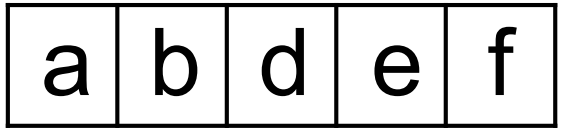
numElements: 5



Push(g)

Stack Implementation with Array

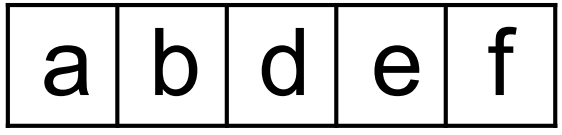
numElements: 5



Push(g) → ERROR

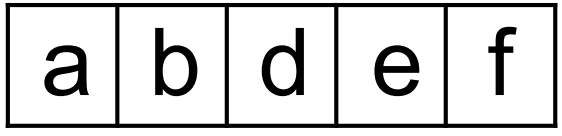
Stack Implementation with Array

numElements: 5



Stack Implementation with Array

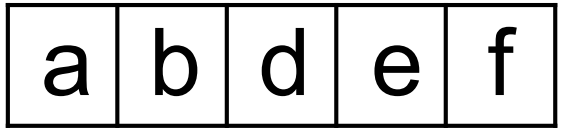
numElements: 5



Empty()

Stack Implementation with Array

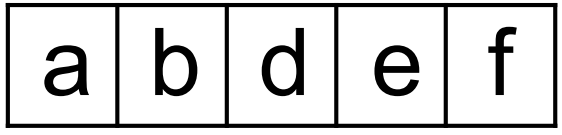
numElements: 5



`Empty()` → `False`

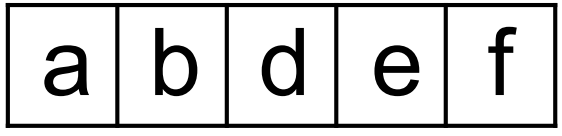
Stack Implementation with Array

numElements: 5



Stack Implementation with Array

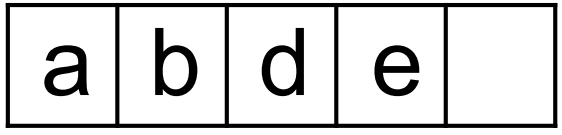
numElements: 5



Pop()

Stack Implementation with Array

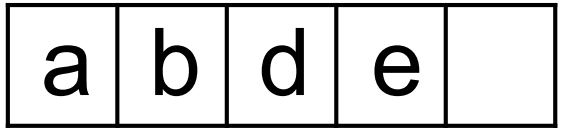
numElements: 4



$\text{Pop}() \rightarrow \text{f}$

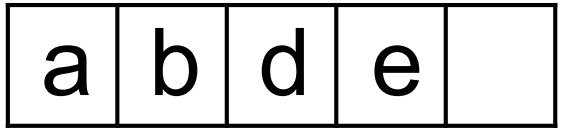
Stack Implementation with Array

numElements: 4



Stack Implementation with Array

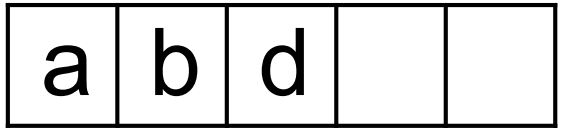
numElements: 4



Pop()

Stack Implementation with Array

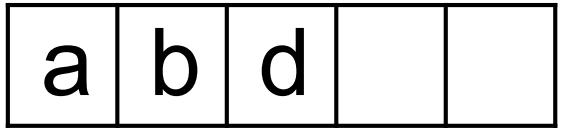
numElements: 3



$\text{Pop}() \rightarrow \text{e}$

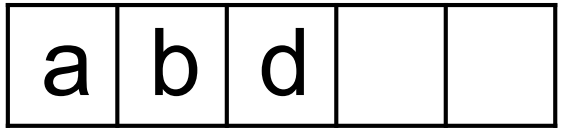
Stack Implementation with Array

numElements: 3



Stack Implementation with Array

numElements: 3



Pop()

Stack Implementation with Array

numElements: 2

$\text{Pop}() \rightarrow \text{d}$

Stack Implementation with Array

numElements: 2

Stack Implementation with Array

numElements: 2

Pop()

Stack Implementation with Array

numElements: 1

$\text{Pop}() \rightarrow \text{b}$

Stack Implementation with Array

numElements: 1

Stack Implementation with Array

numElements: 1

Pop()

Stack Implementation with Array

numElements: 0

$\text{Pop}() \rightarrow a$

Stack Implementation with Array

numElements: 0

Stack Implementation with Array

numElements: 0

Empty()

Stack Implementation with Array

numElements: 0

`Empty()` → `True`

Stack Implementation with Array

numElements: 0

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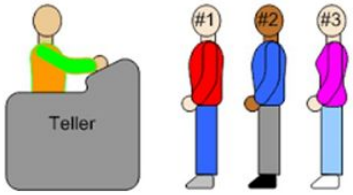
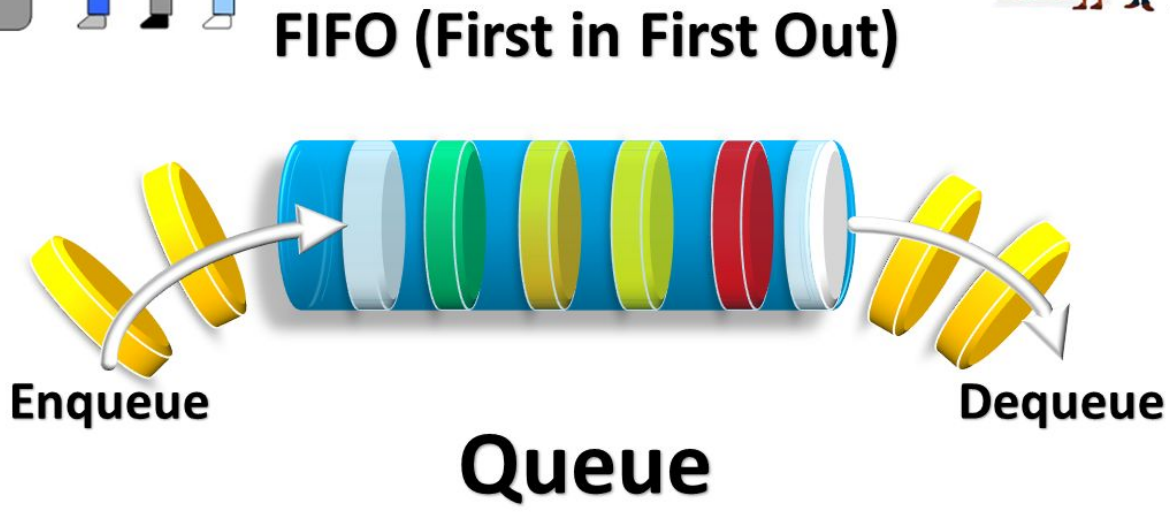
Outline



1. Stacks



2. Queues

Colas

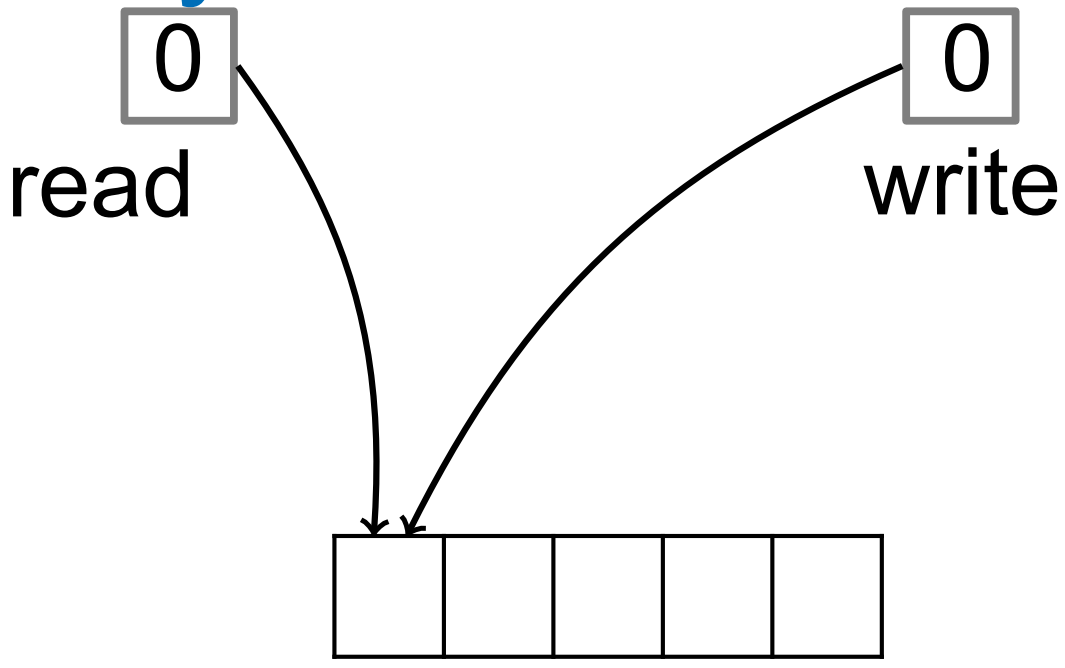
Definition

Queue: Abstract data type with the following operations:

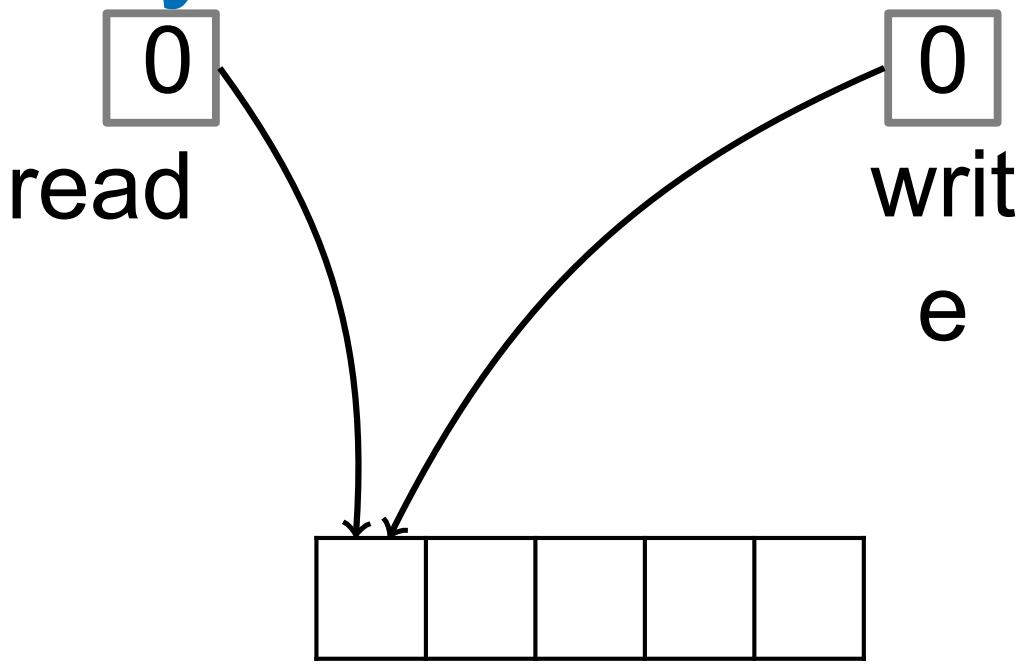
- `Enqueue(key)`: adds key to collection `key Dequeue()`: removes and returns least recently-added key
- `Boolean Empty()`: are there any elements?

FIFO: First-In, First-Out

Queue Implementation with Array

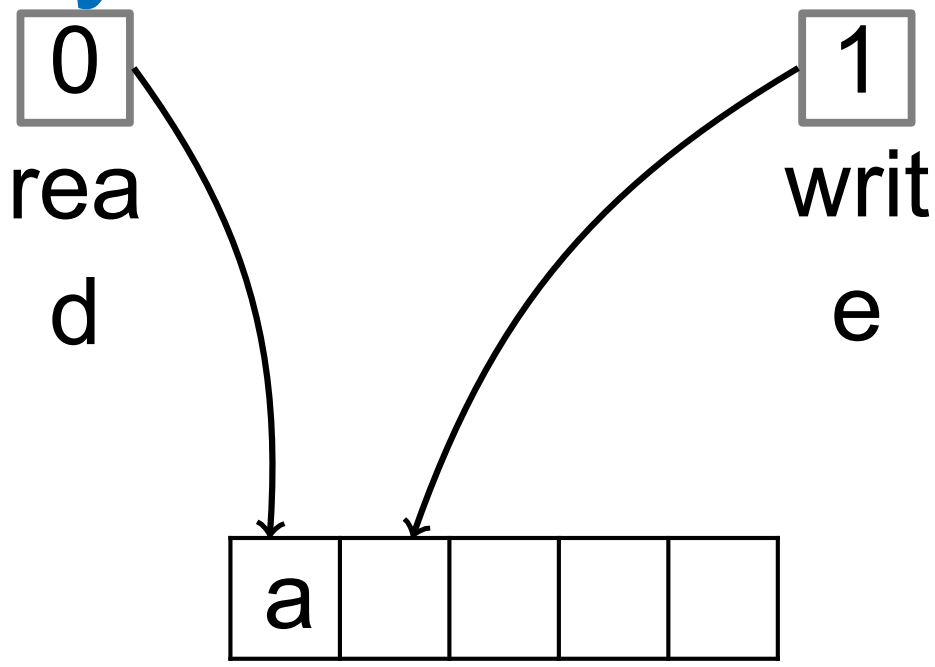


Queue Implementation with Array



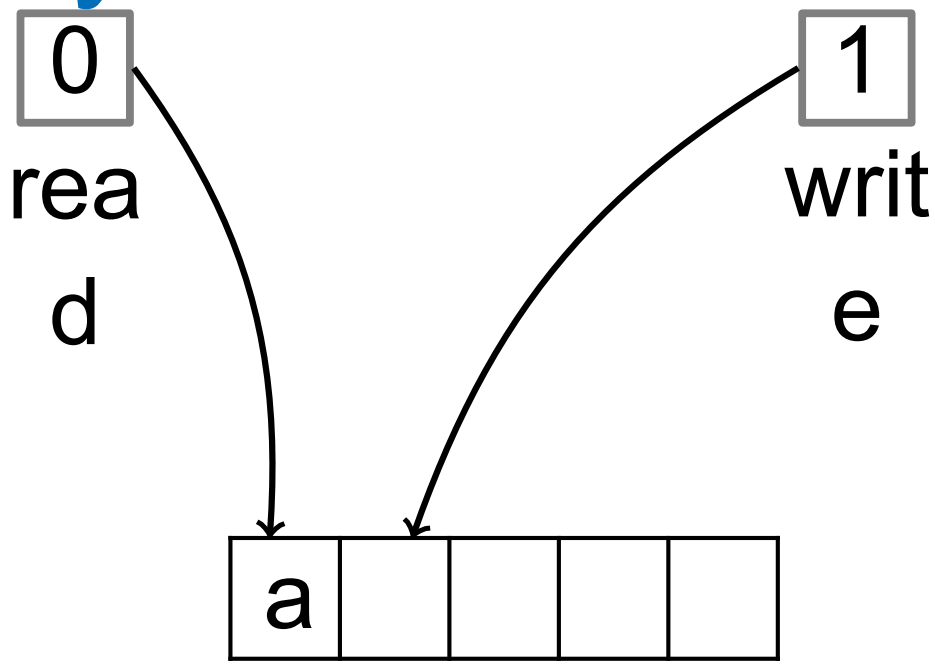
Enqueue(a)

Queue Implementation with Array

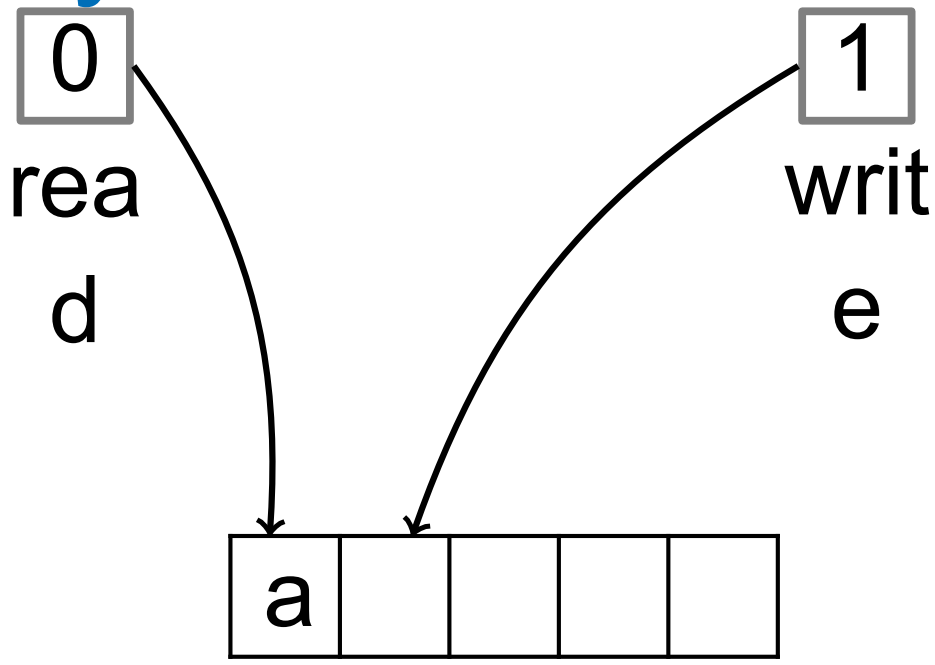


Enqueue(a)

Queue Implementation with Array

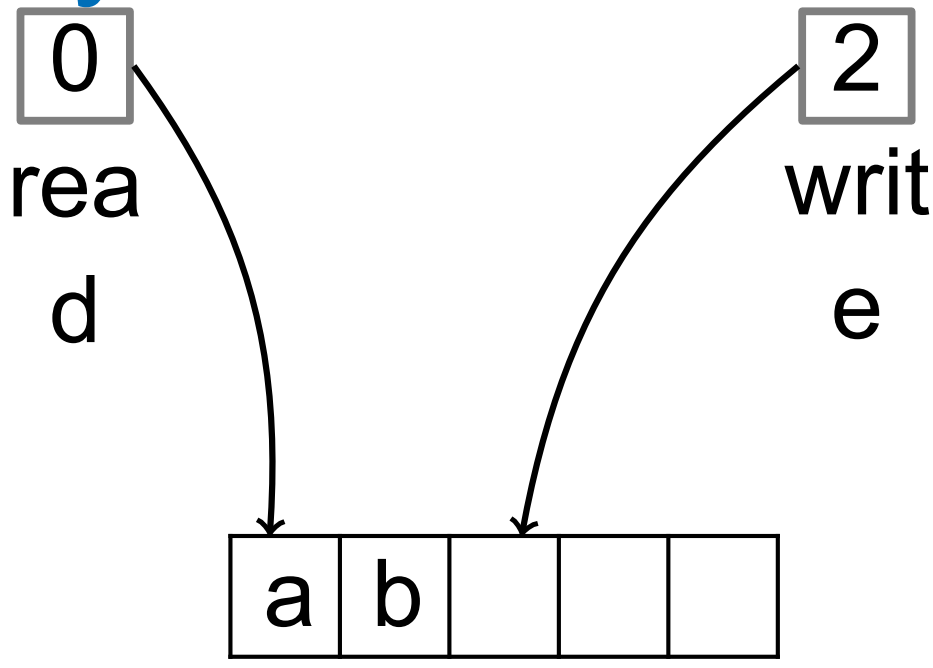


Queue Implementation with Array



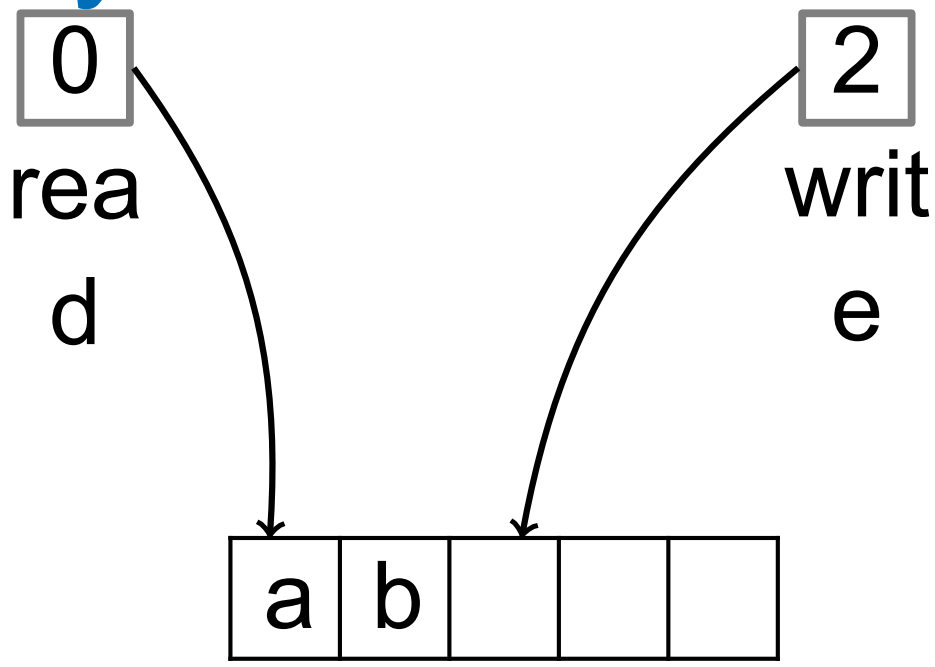
Enqueue (b)

Queue Implementation with Array

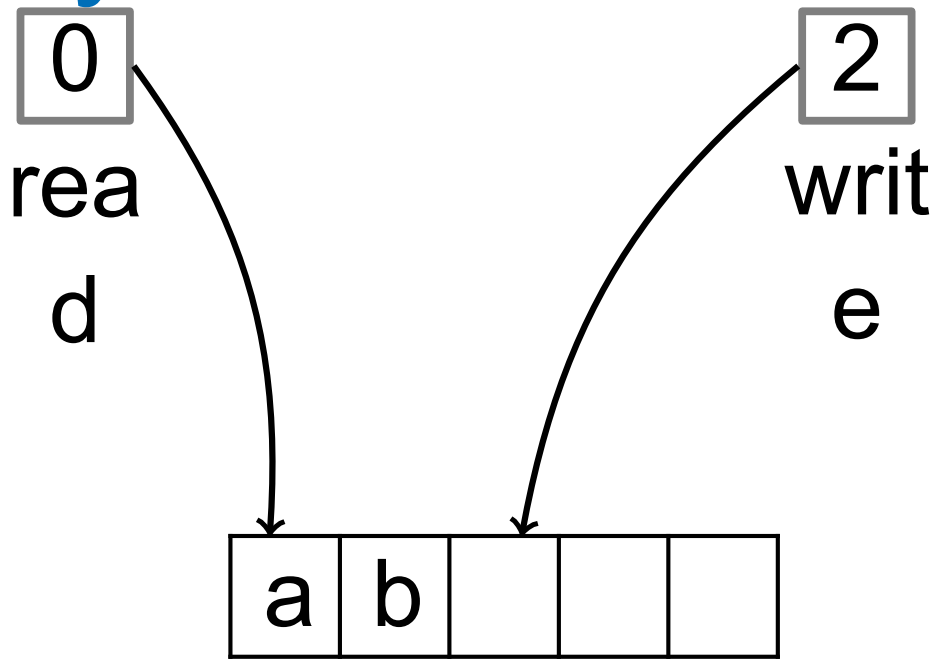


Enqueue (b)

Queue Implementation with Array

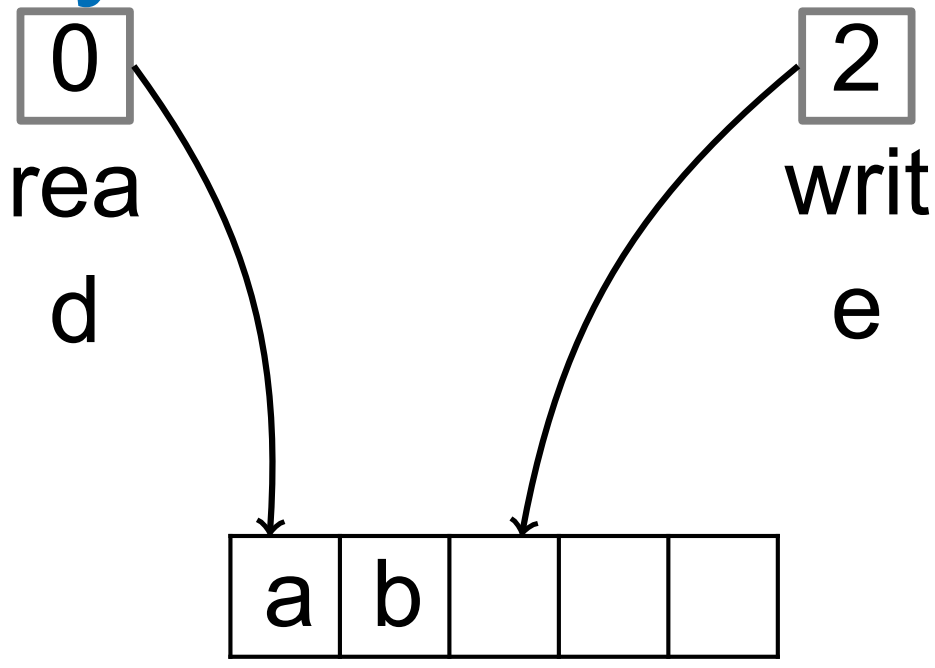


Queue Implementation with Array



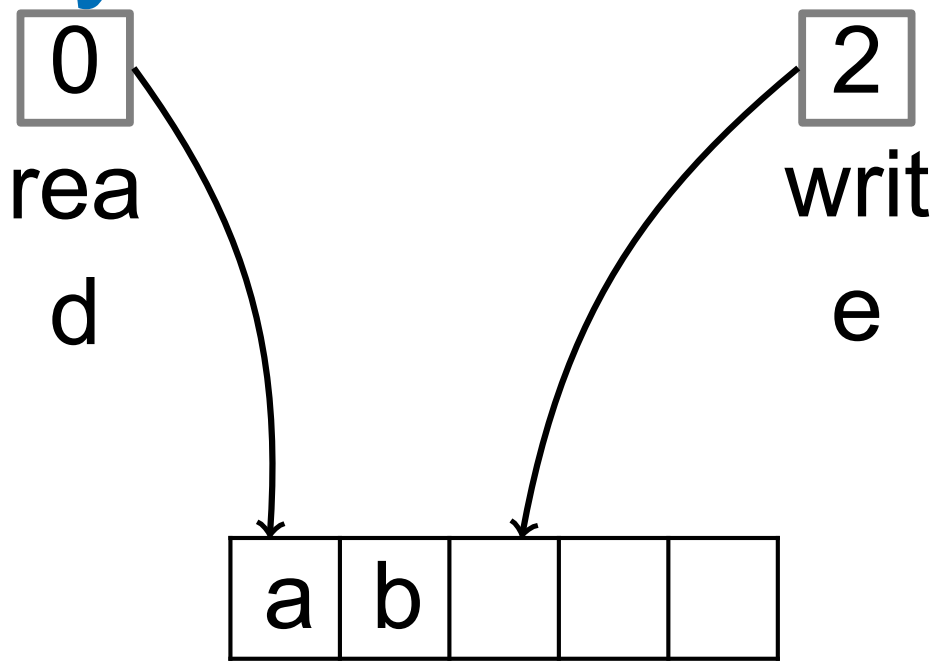
Empty()

Queue Implementation with Array

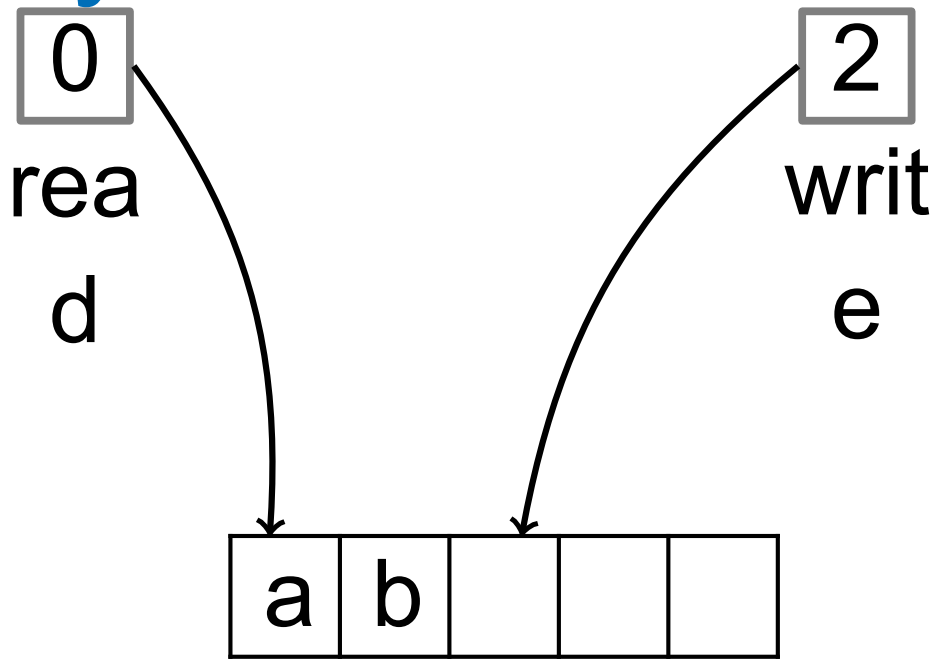


Empty() → False

Queue Implementation with Array

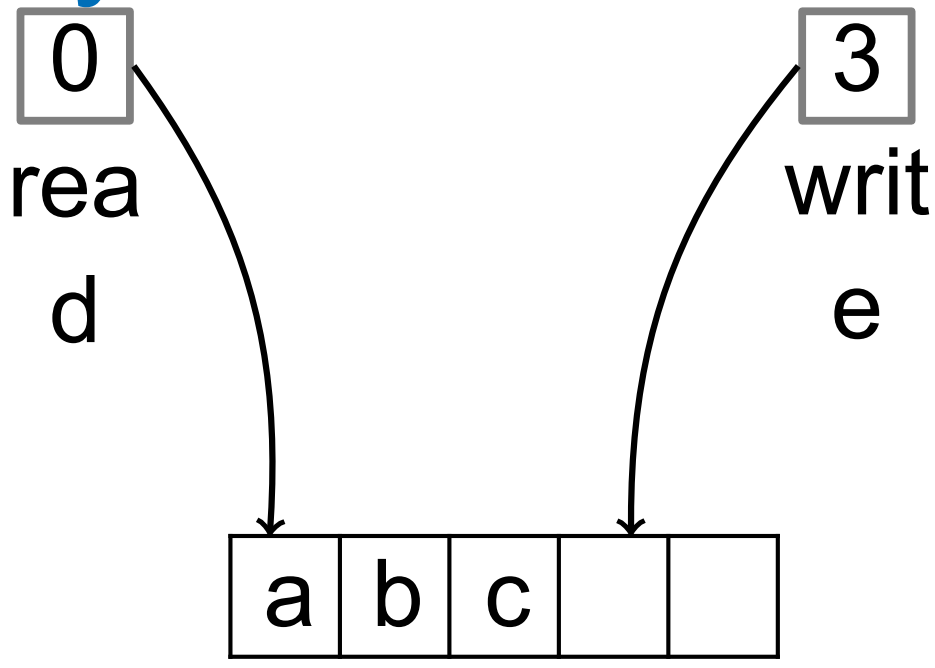


Queue Implementation with Array



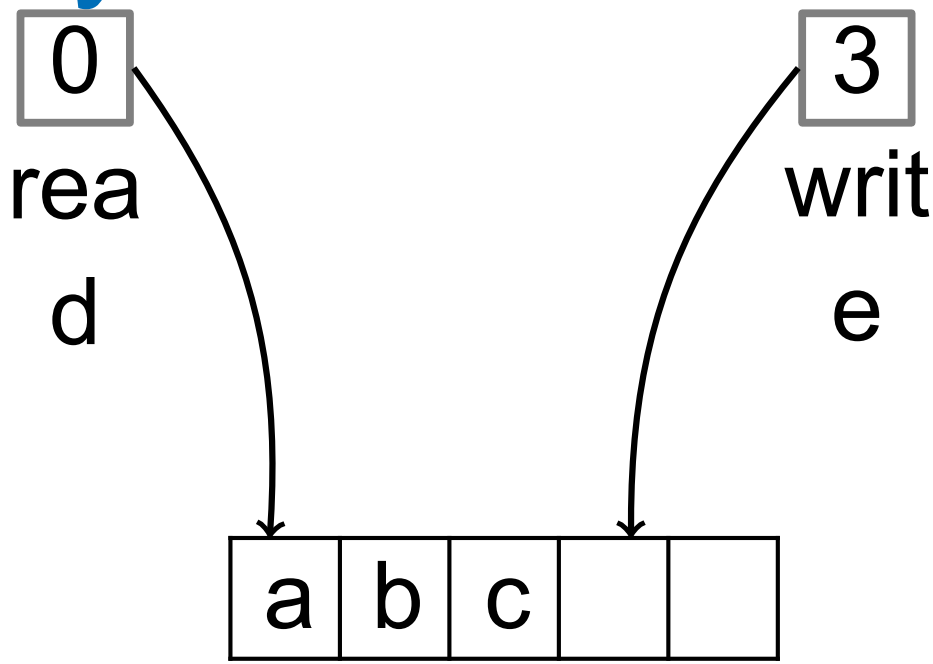
Enqueue(c)

Queue Implementation with Array

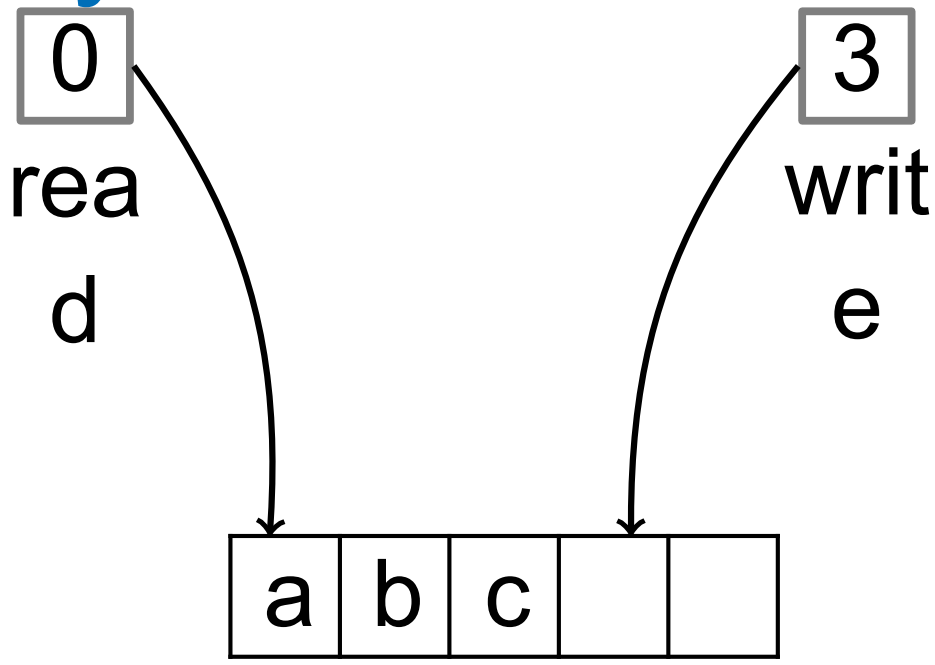


Enqueue(c)

Queue Implementation with Array

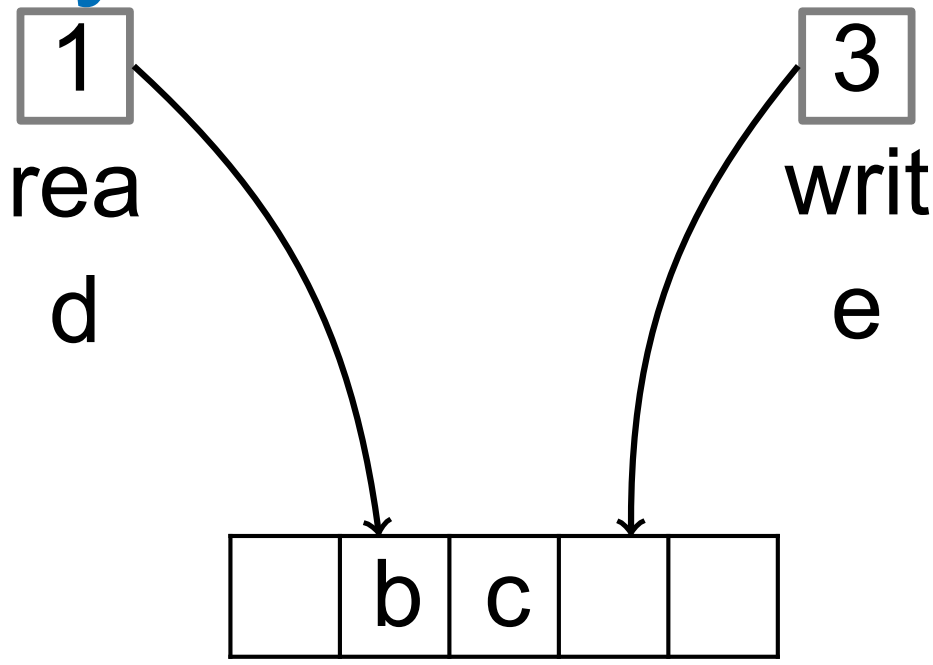


Queue Implementation with Array



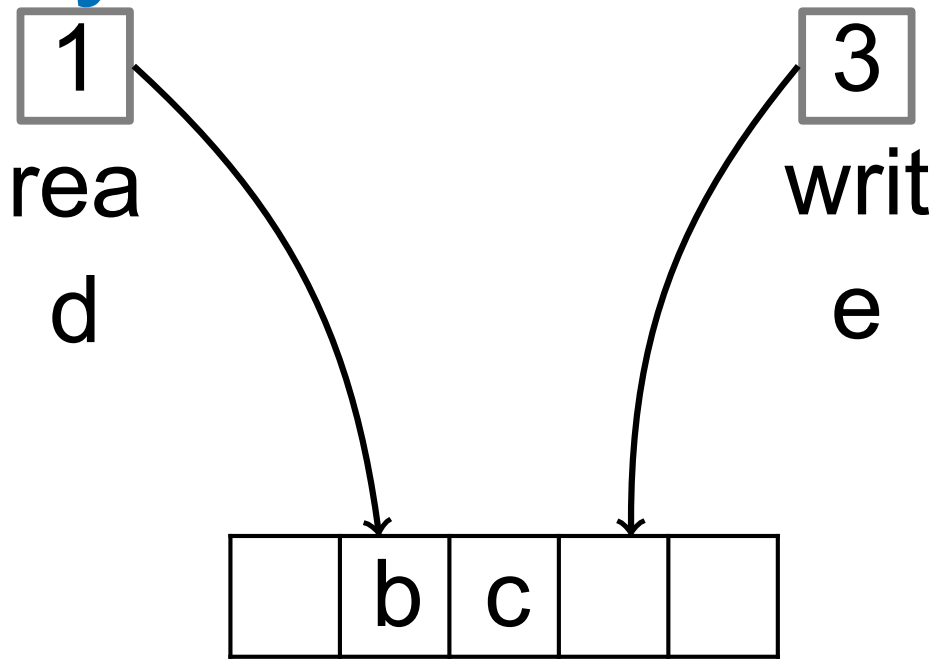
Deąueue()

Queue Implementation with Array

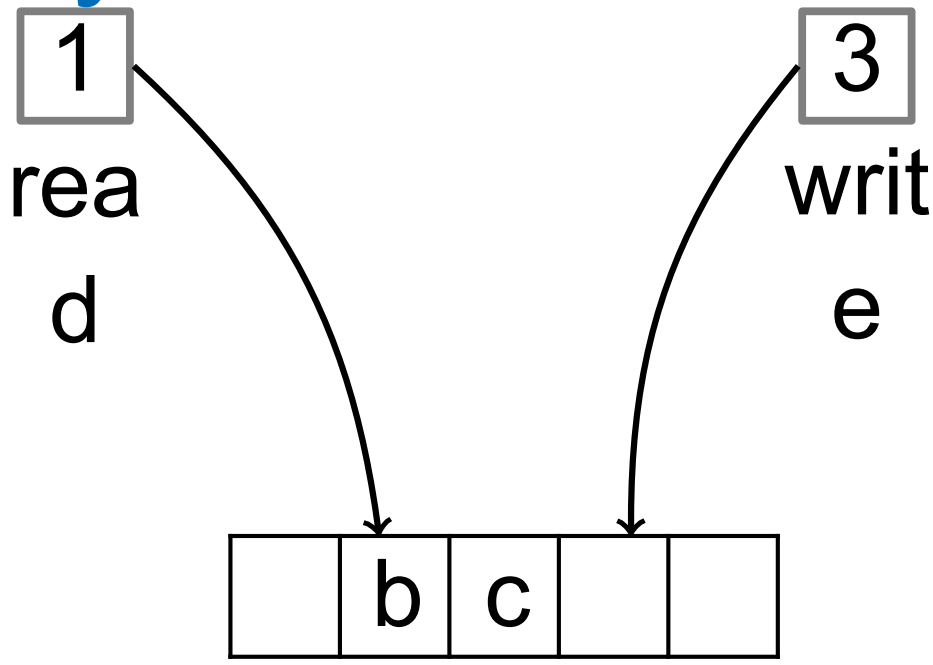


Dequeue() → a

Queue Implementation with Array

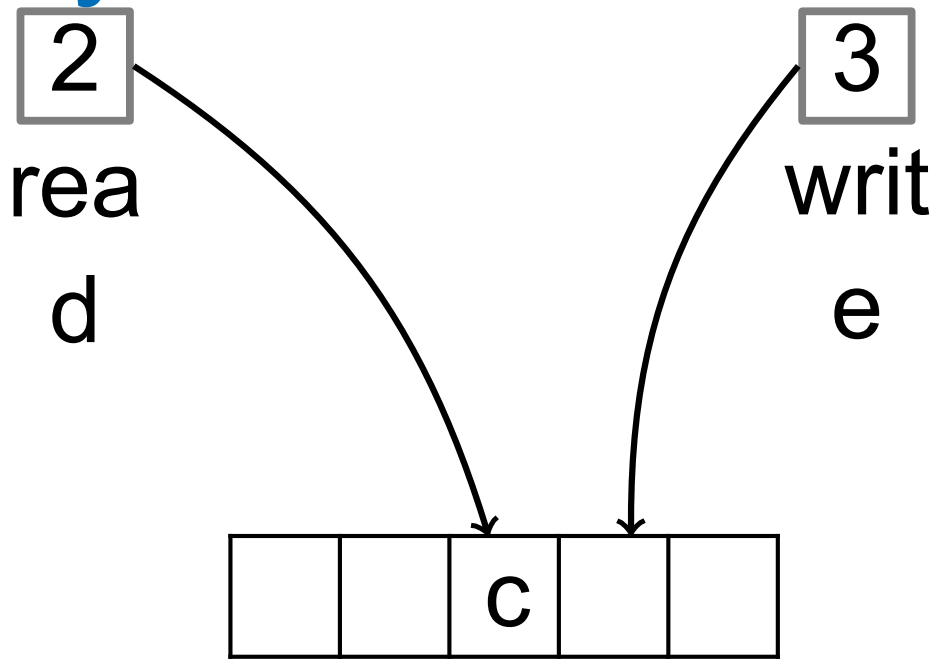


Queue Implementation with Array



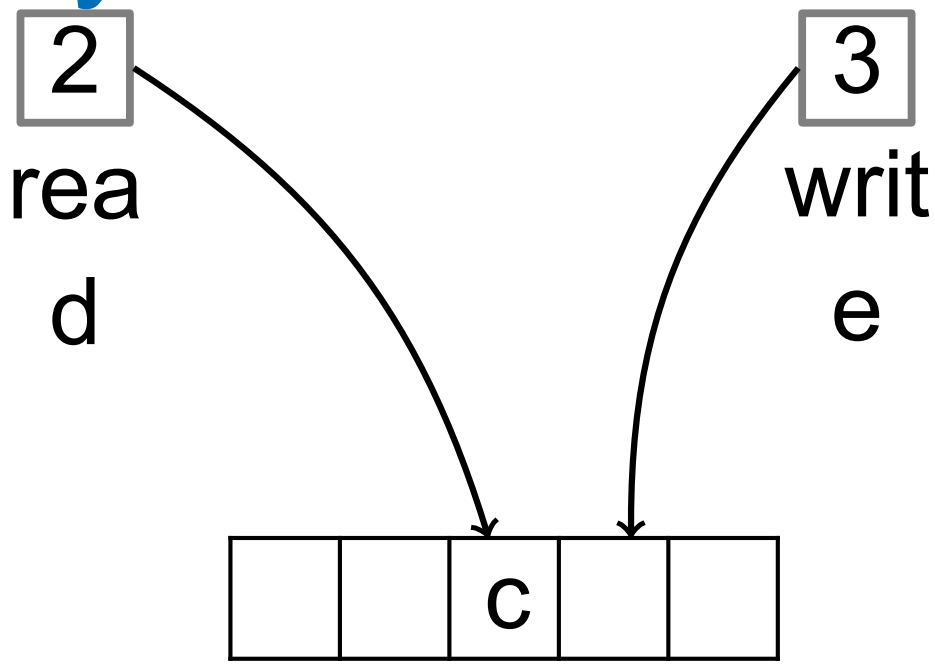
Deąueue()

Queue Implementation with Array

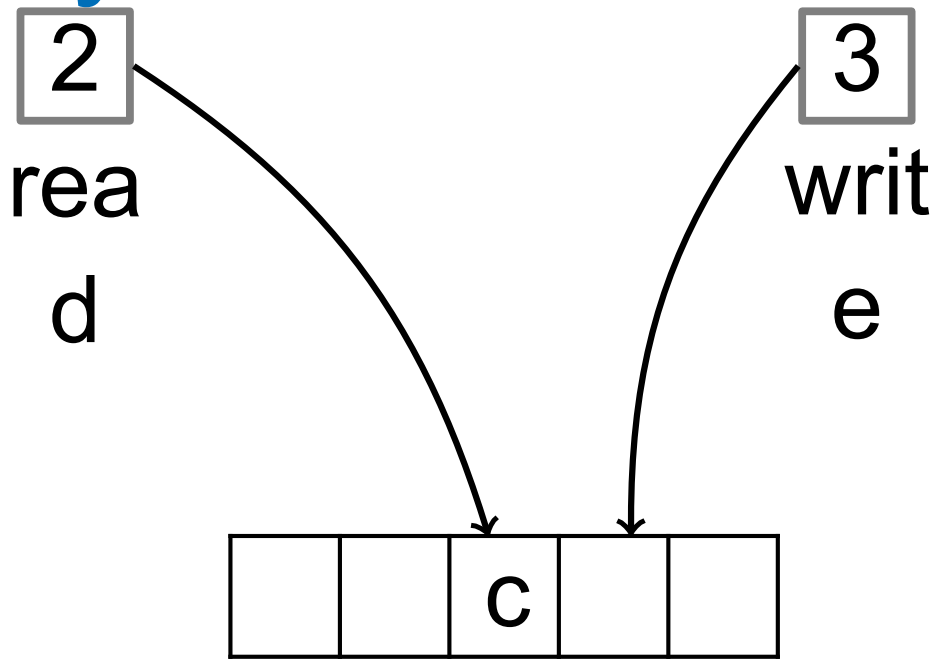


Deąueue () → b

Queue Implementation with Array

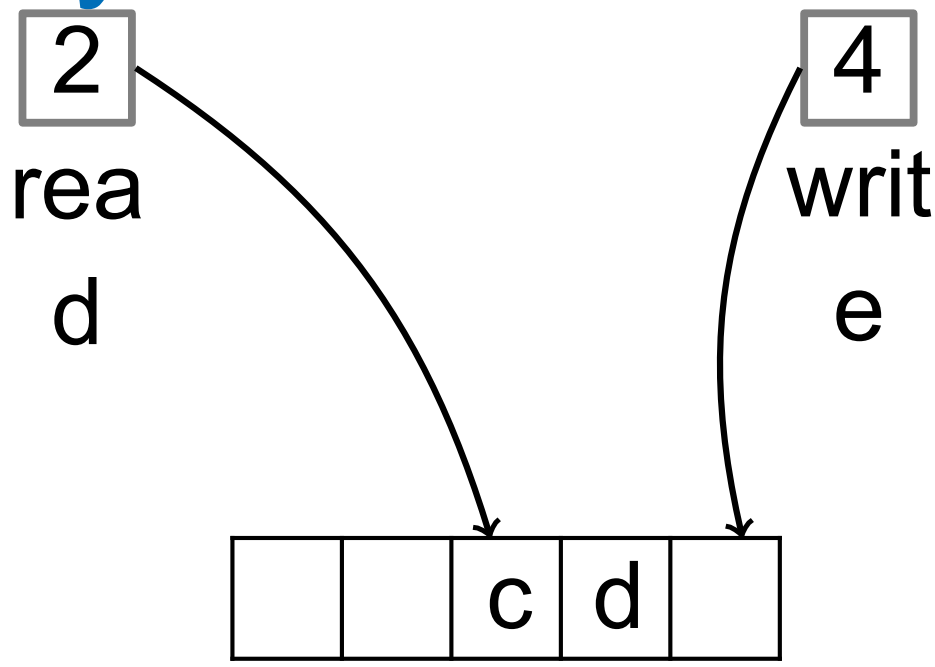


Queue Implementation with Array



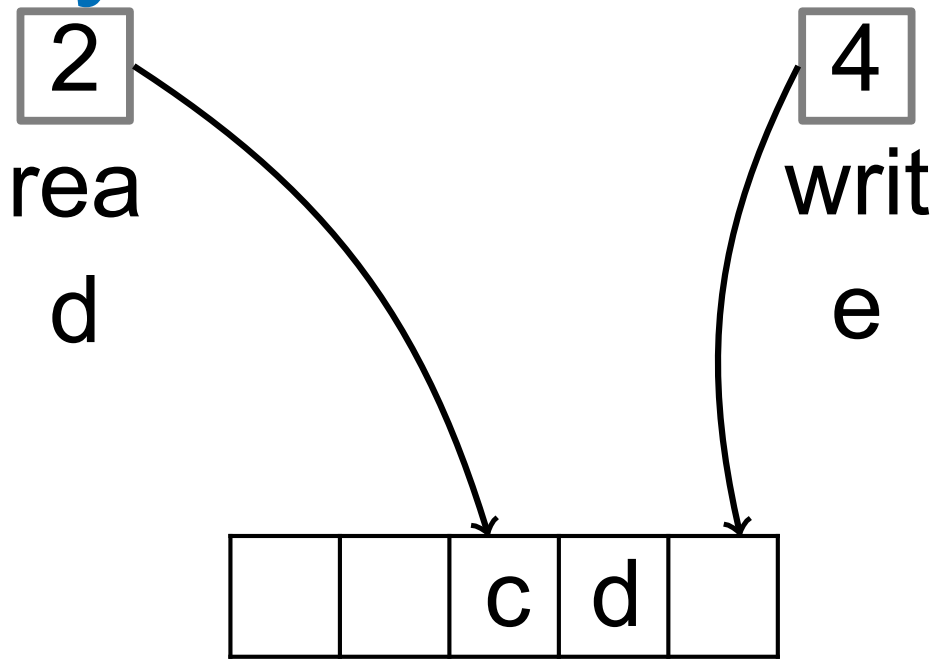
Enqueue(d)

Queue Implementation with Array

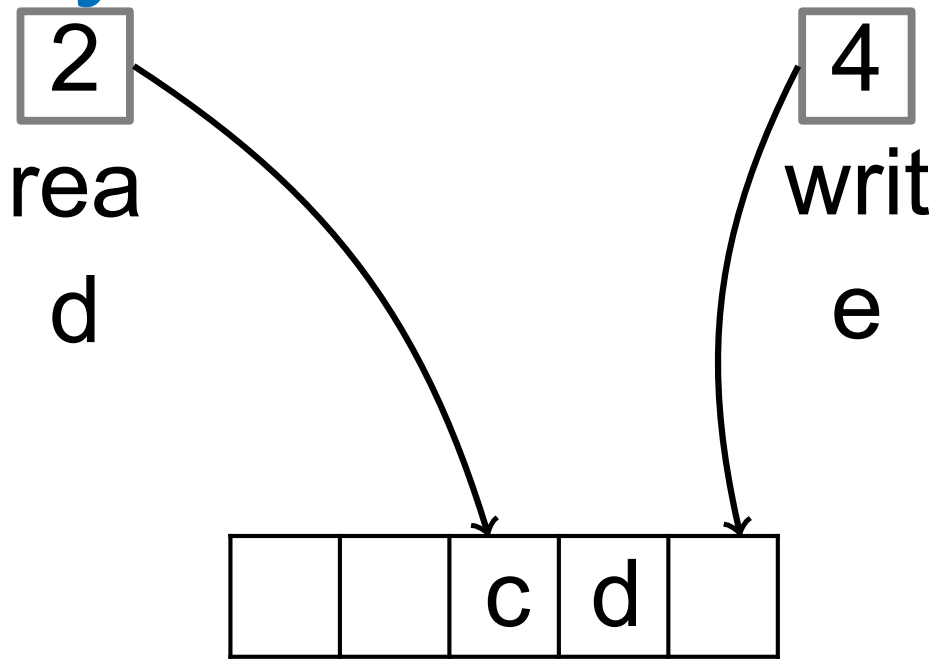


Enqueue(d)

Queue Implementation with Array

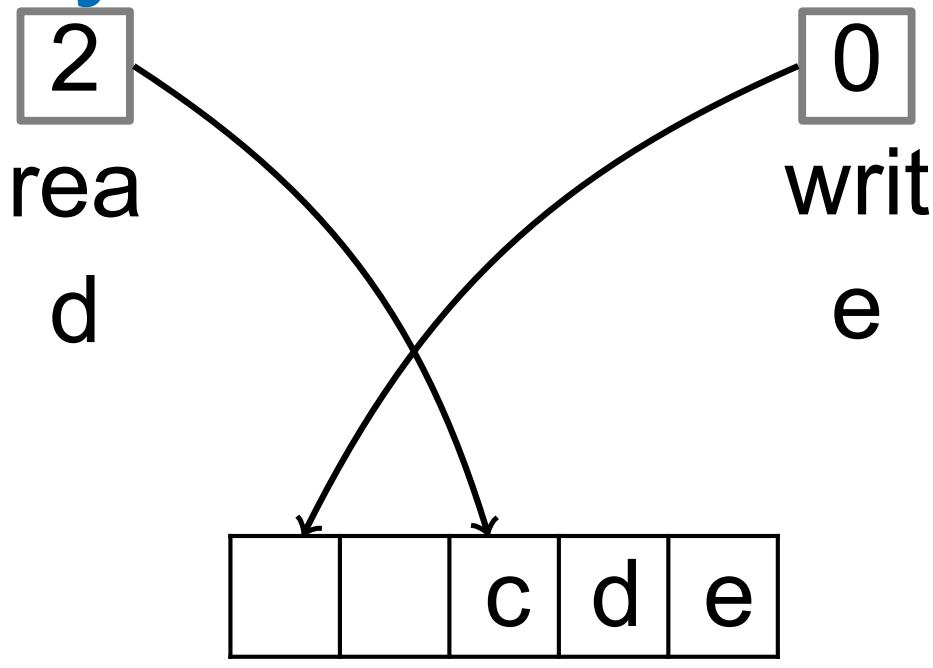


Queue Implementation with Array



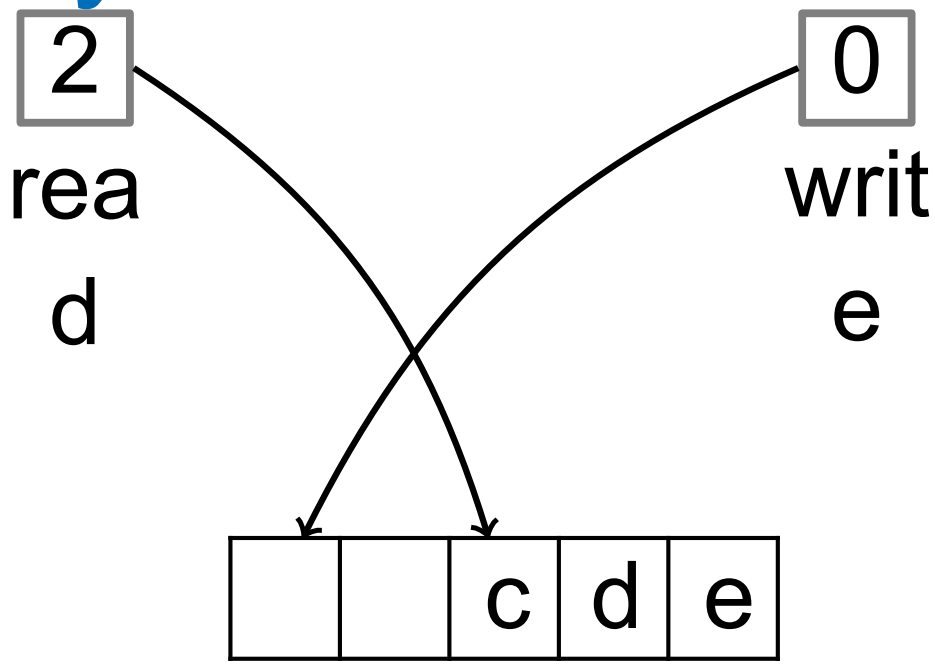
Enqueue(e)

Queue Implementation with Array

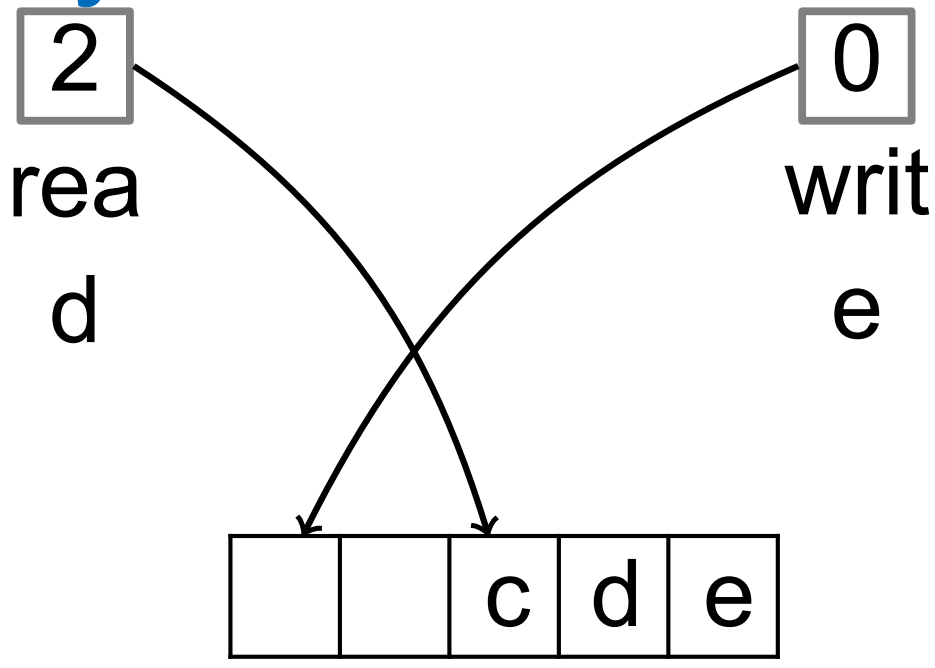


Enqueue(e)

Queue Implementation with Array

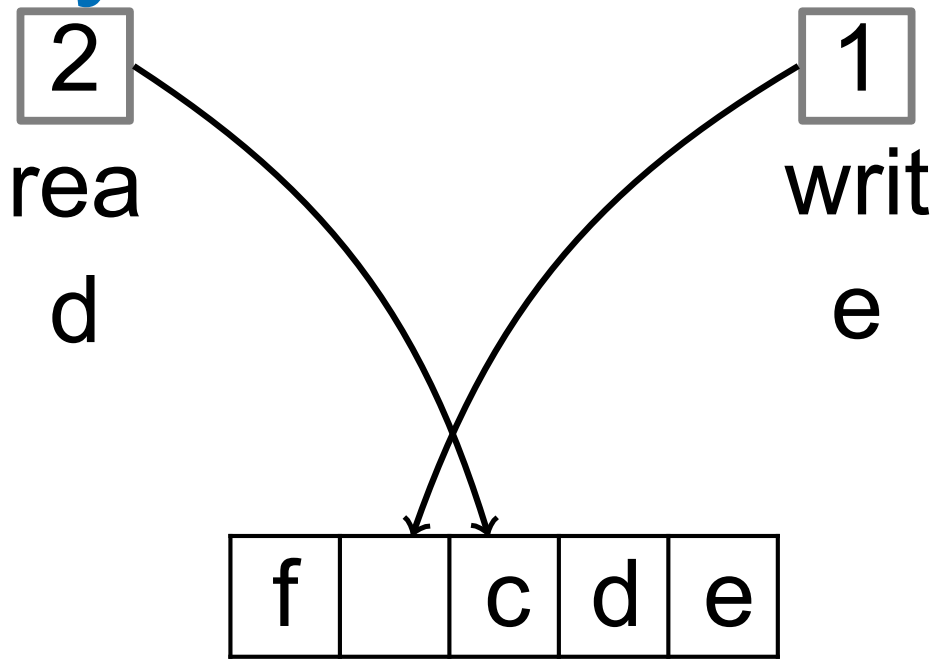


Queue Implementation with Array



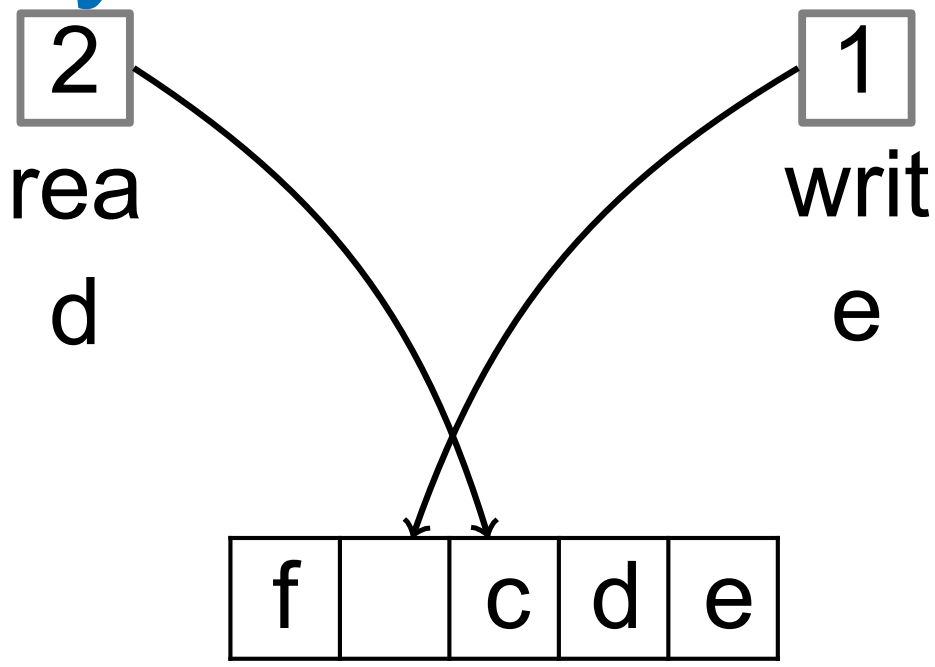
Enqueue(f)

Queue Implementation with Array

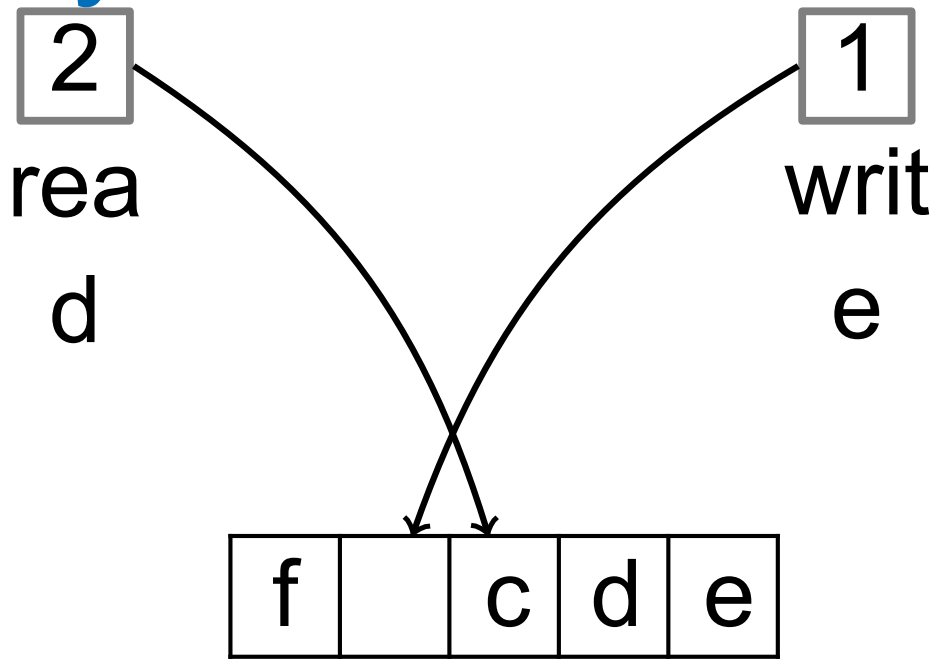


Enqueue(f)

Queue Implementation with Array

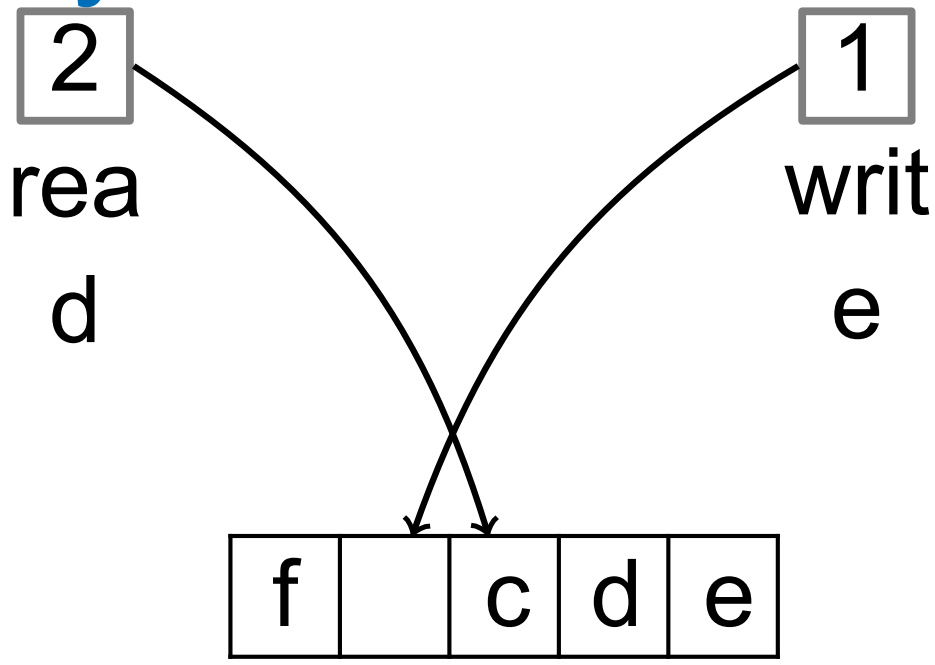


Queue Implementation with Array



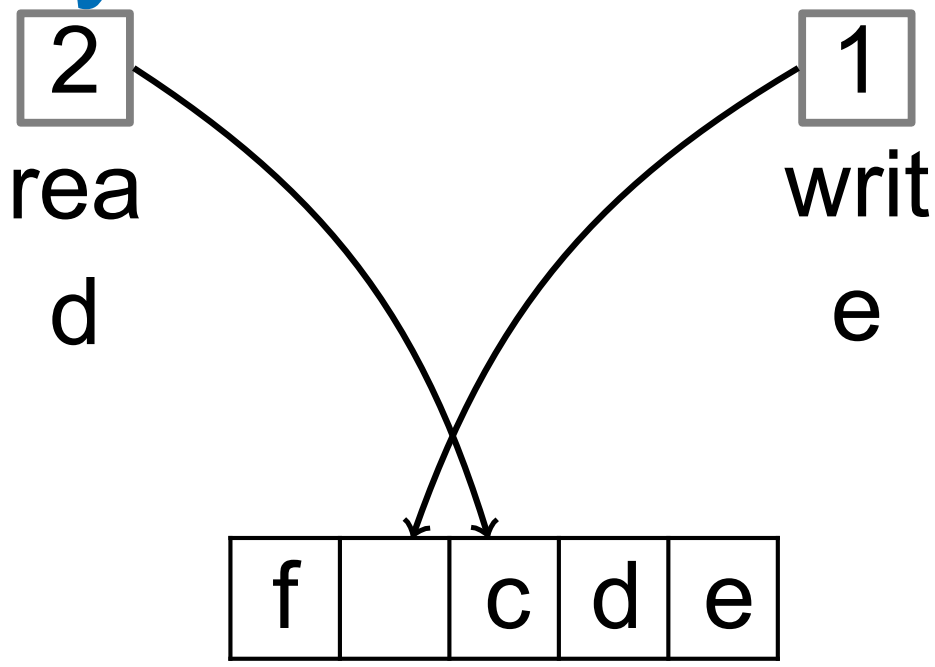
Enqueue(g)

Queue Implementation with Array

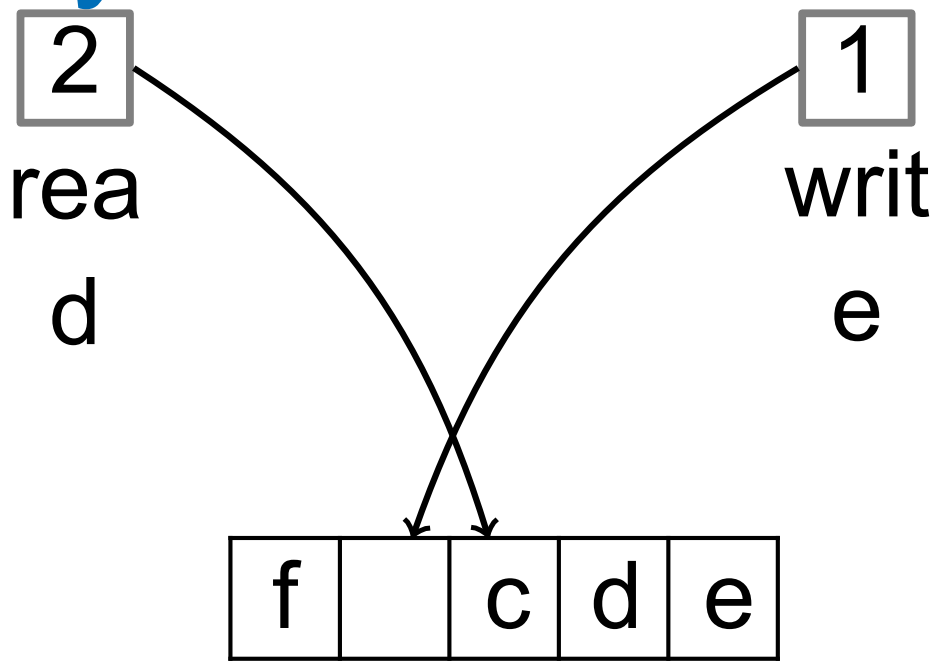


Enqueue(g) → ERROR

Queue Implementation with Array

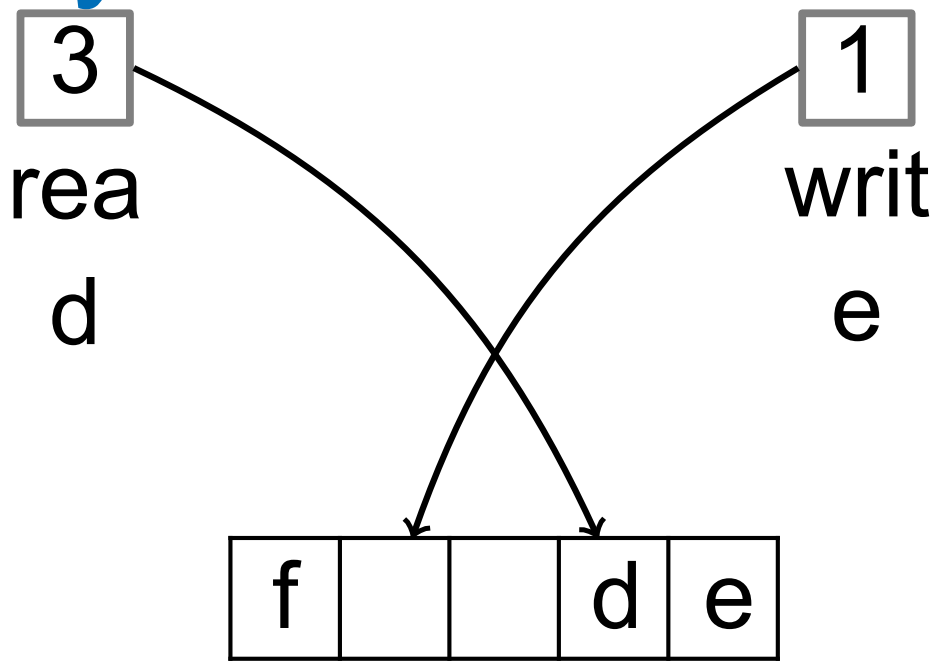


Queue Implementation with Array



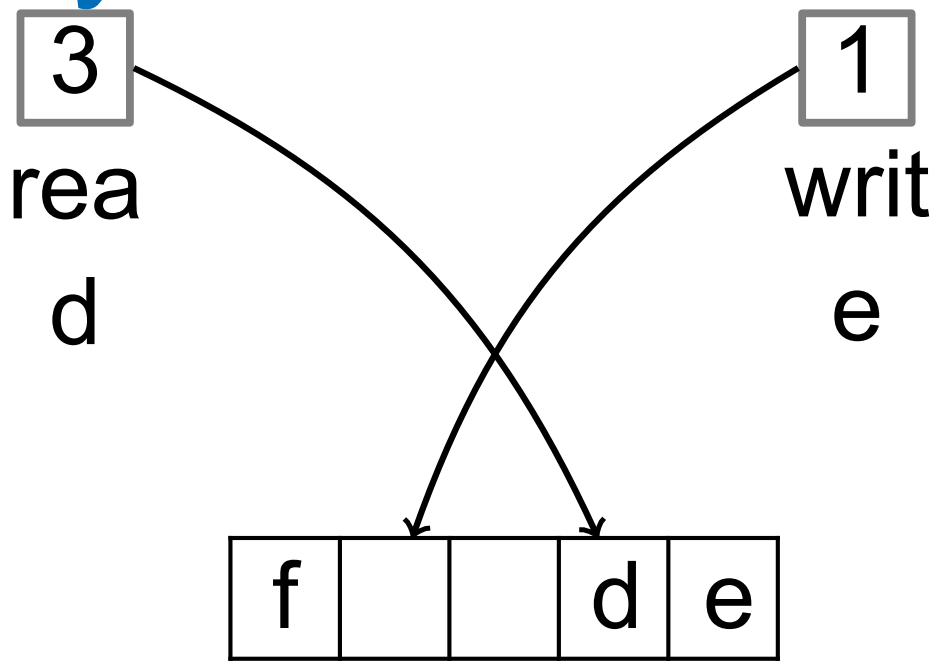
Dequeue()

Queue Implementation with Array

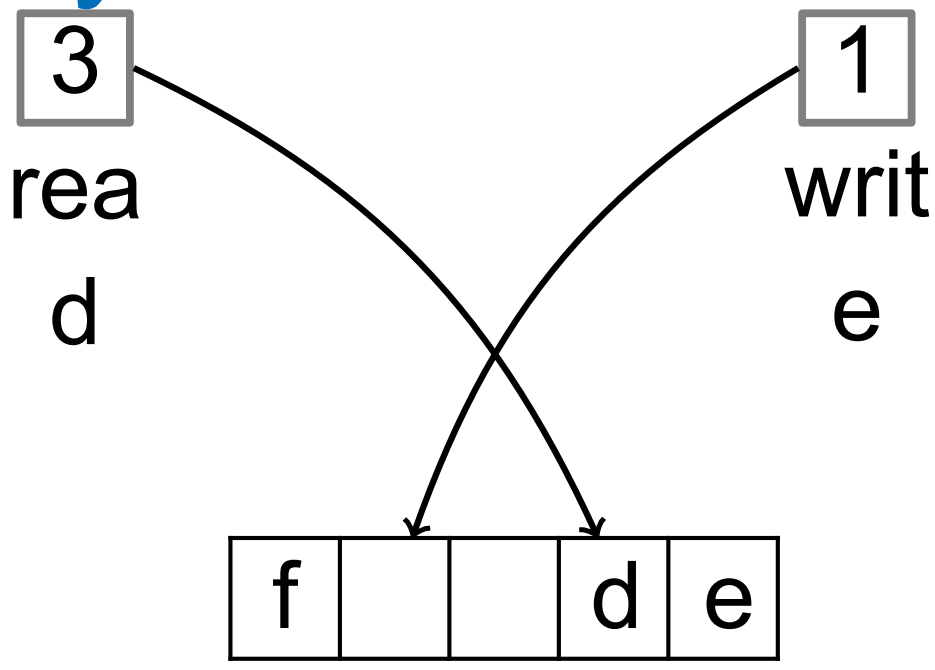


Deąueue () → c

Queue Implementation with Array

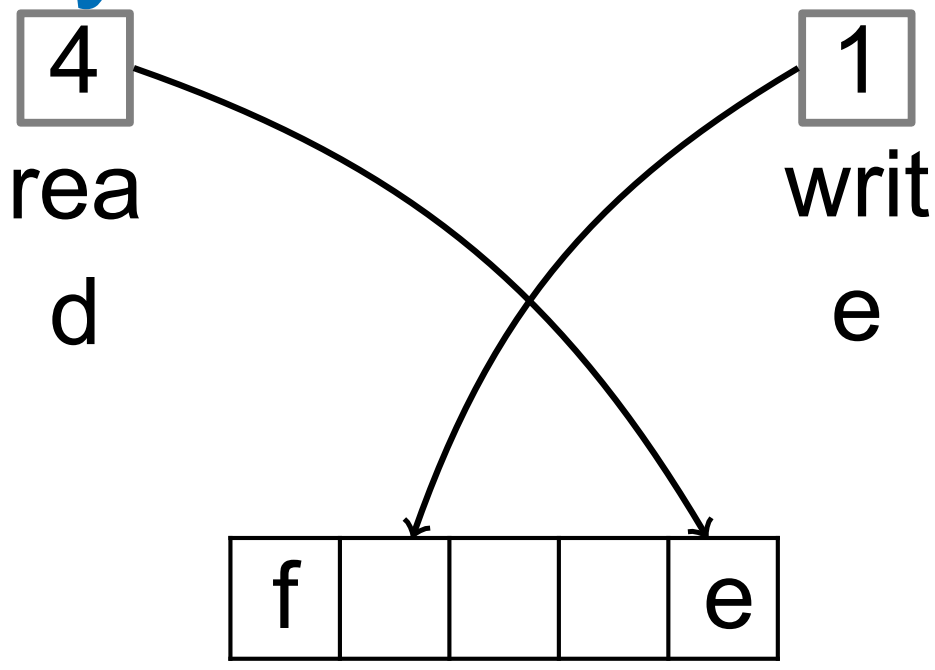


Queue Implementation with Array



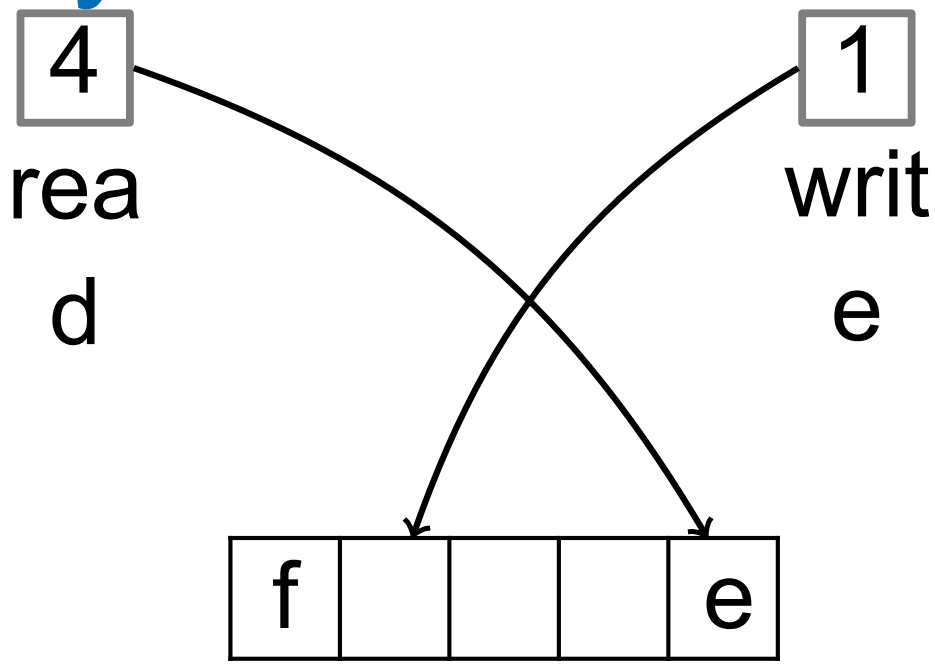
Deąueue()

Queue Implementation with Array

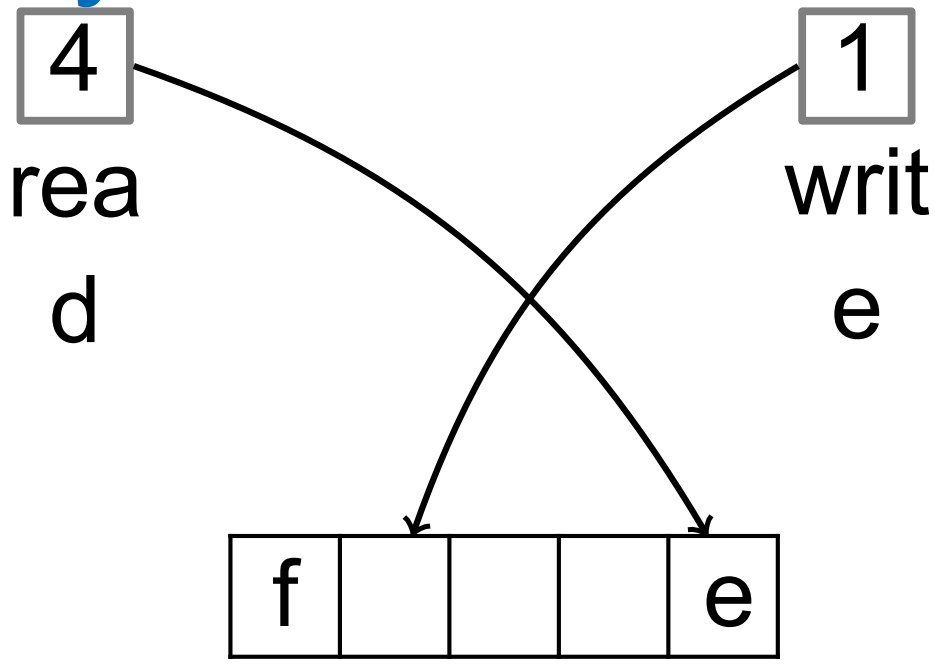


Deąueue () → d

Queue Implementation with Array

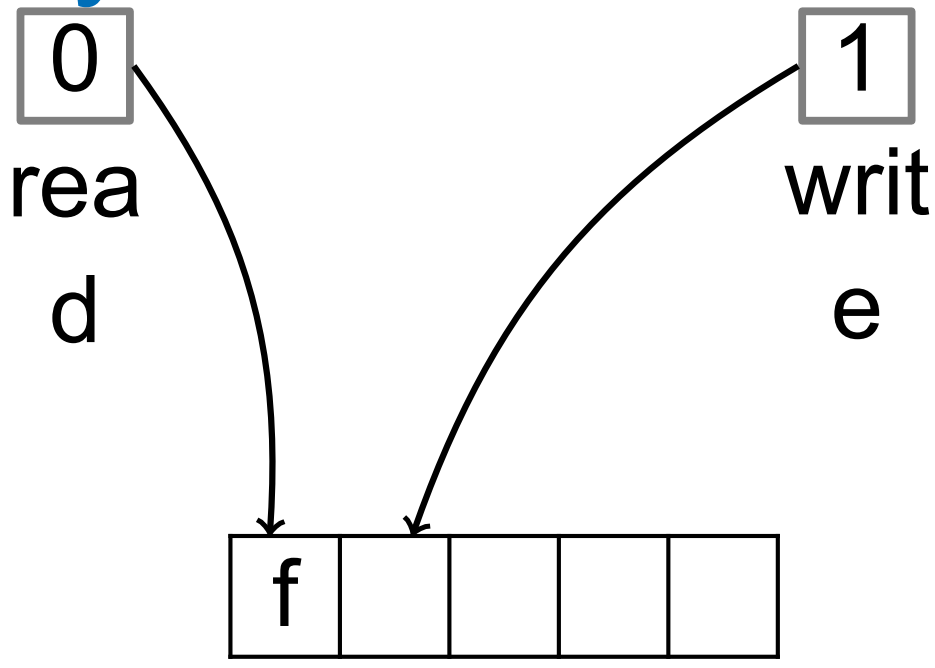


Queue Implementation with Array



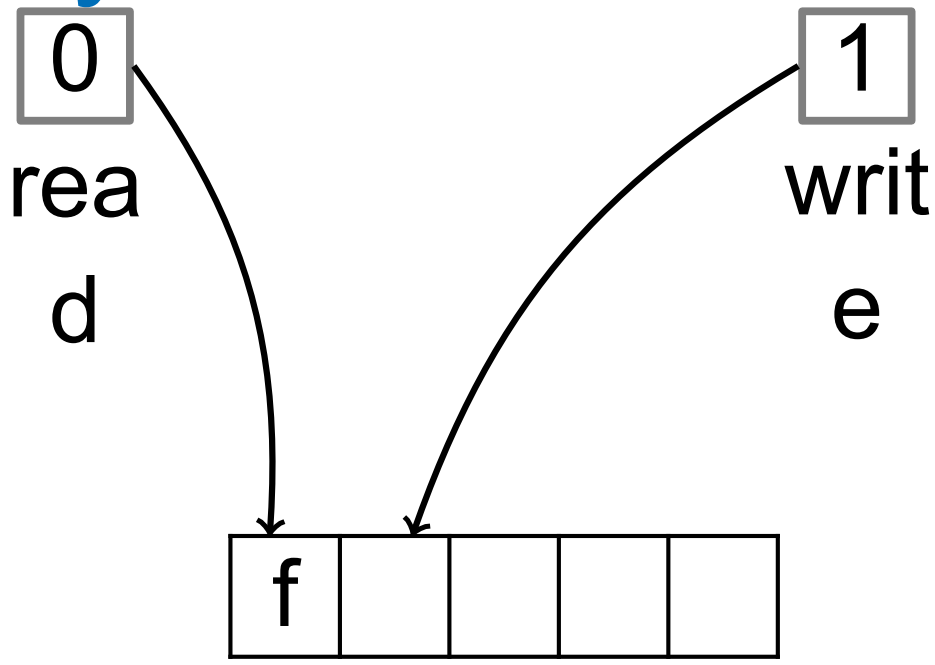
Deąueue()

Queue Implementation with Array

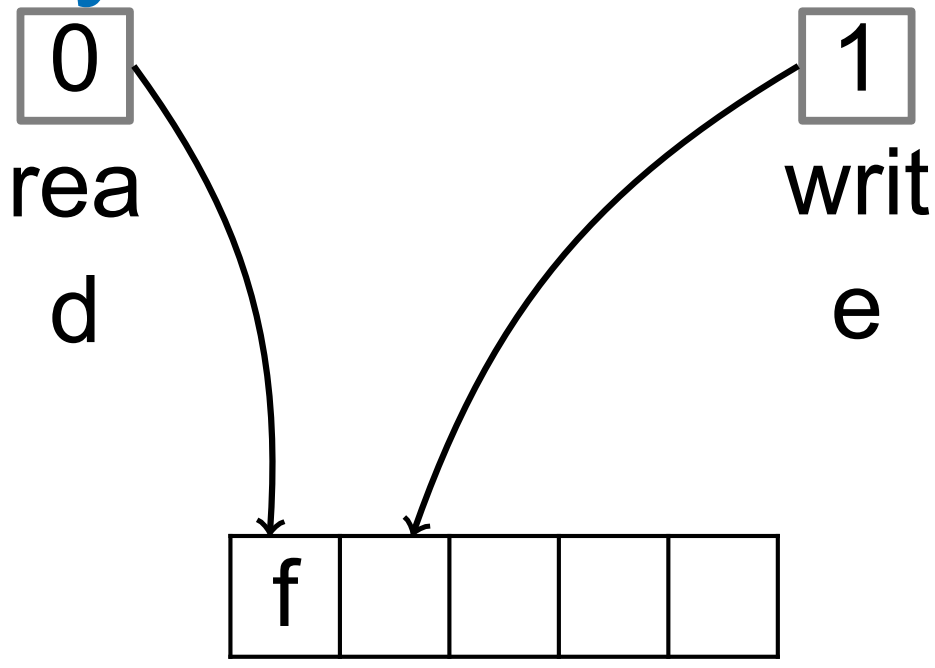


Deąueue () → e

Queue Implementation with Array

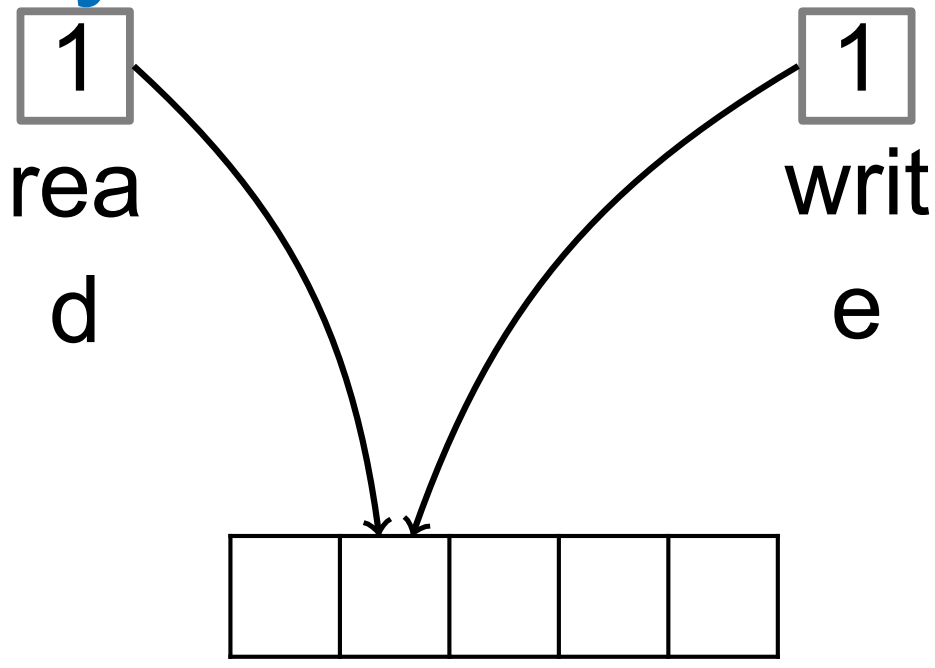


Queue Implementation with Array



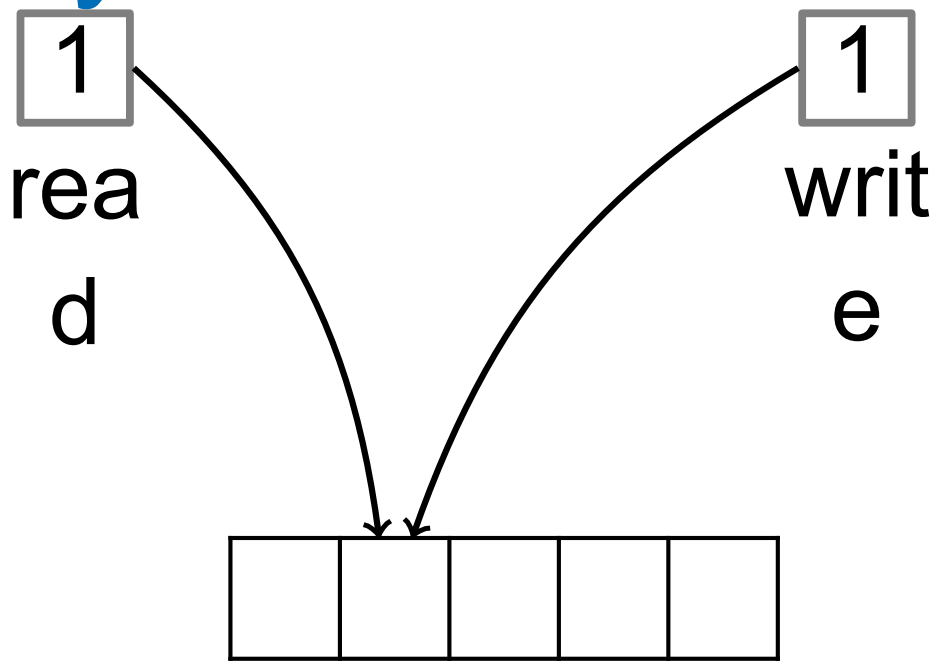
Deąueue()

Queue Implementation with Array

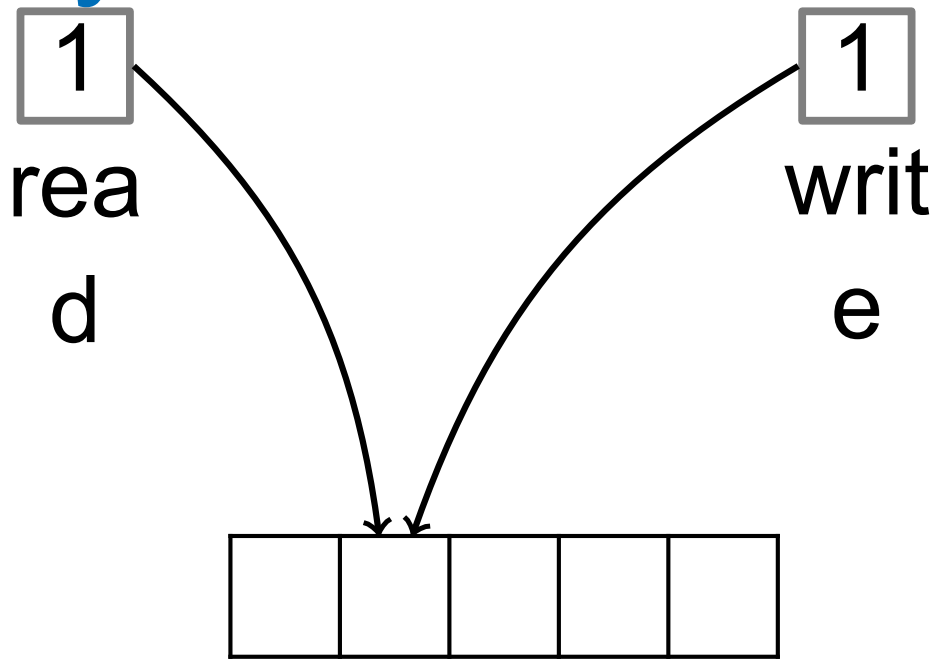


Dequeue() → f

Queue Implementation with Array

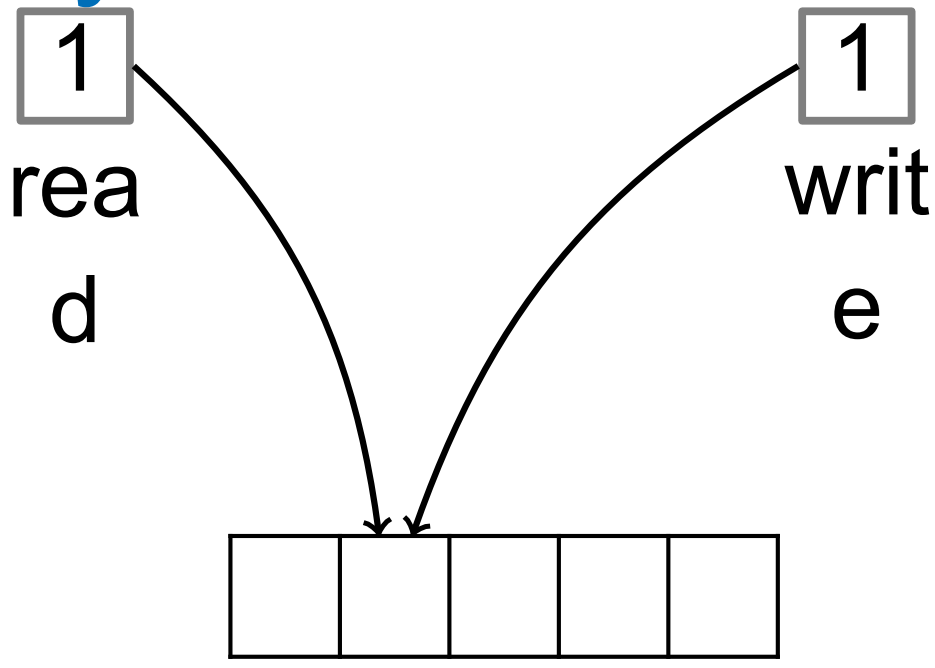


Queue Implementation with Array



Empty()

Queue Implementation with Array



Empty() → True

Summary

- Queues can be implemented with either a linked list (with tail pointer) or an array.
- Each queue operation is $O(1)$: Enqueue, Dequeue, Empty.

https://sites.google.com/unal.edu.co/data/